

i 2024 Front page

Department of Engineering Cybernetics

Examination paper for TTK4215 Adaptive Control

Date: 03.12.2024

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Present at the exam location: NO

Permitted examination support material: D / No support material is allowed. A specific basic calculator is allowed.

OTHER INFORMATION

Read the questions carefully and make your own assumptions. Specify in your answer which assumptions you have used as a basis for interpreting/defining the assignment.

The academic person is only contacted in case of errors or insufficiencies in the question set. Address an invigilator if you suspect errors or insufficiencies. Write down the question in advance.

SPECIFIC INFORMATION FOR YOUR COURSE

Hand drawings: For question **MRAC c) and d)** you are meant to answer on handwritten sheets. Other questions must be answered directly in Inspira. At the bottom of the question, you will find a seven-digit code. Fill in this code in the top left corner of the sheets you wish to submit.

We recommend that you do this during the exam. If you require access to the codes after the examination time ends, click "Show submission".

You are responsible for filling in the correct codes on the handwritten sheets. Therefore, read the cover sheet carefully. The Examination Office cannot guarantee that incorrectly completed sheets will be added to your assignment.

Notifications: Any messages during the exam (e.g., in case of errors in the exam set) will be sent out via notifications in Inspira. A notification will appear as a dialog box on the screen. You can find the notification again by clicking on the bell icon at the top right.

Withdrawing from the exam: If you wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click "Submit blank". This cannot be undone, even if the test is still open.

Access to your answers: After the exam, you can find your answers under previous tests in Inspira. Be aware that it may take a working day until any hand-written material is available in "previous tests".

1 2024 - Barbalat's Lemma

Specify whether the claims below are true or false (wrong answer gives negative score, while "Don't know" gives 0 points).

If $g \in \mathcal{L}_2 \cap \mathcal{L}_\infty$ then $g(t)$ converges to 0 when $t \rightarrow \infty$.

☐ True

☐ False



☐ Don't know

Let $g(t) = \frac{\sin(t^{1/4})}{1+t}$, $t \in [0, \infty)$.

$g \in \mathcal{L}_\infty$

☐ True



☐ False

☐ Don't know

$g \in \mathcal{L}_2$

☐ True



☐ False

☐ Don't know

$\dot{g} \in \mathcal{L}_\infty$

☐ True

☐ False



☐ Don't know

$\lim_{t \rightarrow \infty} g(t) = 0$

- ☐ True
- ☐ False
- ☐ Don't know



Maks poeng: 5

2 2024 Gradient method

Consider the linear parametric model

$$z(t) = \theta^{*T} \phi(t)$$

where $z : [0, \infty) \rightarrow \mathbb{R}$ and $\phi : [0, \infty) \rightarrow \mathbb{R}^n$ are measured and $\theta^* \in \mathbb{R}^n$ is an unknown constant. Let an estimate $\theta(t)$ of θ^* be computed by the gradient method with instantaneous cost, given by

$$\hat{z}(t) = \theta(t)^T \phi(t),$$

$$\epsilon(t) = \frac{z(t) - \hat{z}(t)}{m(t)^2}, m(t)^2 = 1 + n_s(t)^2, n_s(t)^2 = \phi(t)^T \phi(t),$$

$$\dot{\theta}(t) = \Gamma \epsilon \phi,$$

where Γ is a symmetric positive definite matrix. Which of the properties below can you prove by using the Lyapunov function candidate $V(t) = \frac{1}{2} \tilde{\theta}^T \Gamma^{-1} \tilde{\theta}$, where $\tilde{\theta}(t) = \theta(t) - \theta^*$?

Select zero or more alternatives (wrong selections give negative score)

- ☐ $V \in \mathcal{L}_2$
- ☐ $\epsilon n_s \in \mathcal{L}_2$ ✓
- ☐ $\dot{\theta} \in \mathcal{L}_\infty$ ✓
- ☐ $\epsilon \in \mathcal{L}_2$ ✓
- ☐ $\lim_{t \rightarrow \infty} V(t) = V_\infty, V_\infty \in [0, V(0)]$ ✓
- ☐ $\tilde{\theta} \in \mathcal{L}_2$
- ☐ $\hat{z} \in \mathcal{L}_\infty$
- ☐ $z \in \mathcal{L}_2$
- ☐ $\theta \in \mathcal{L}_\infty$ ✓
- ☐ $\theta \in \mathcal{L}_2$
- ☐ $\dot{\theta} \in \mathcal{L}_2$ ✓
- ☐ $\epsilon \in \mathcal{L}_\infty$ ✓
- ☐ $V \in \mathcal{L}_\infty$ ✓
- ☐ $z \in \mathcal{L}_\infty$
- ☐ $\lim_{t \rightarrow \infty} \theta(t) = \theta^*$
- ☐ $\lim_{t \rightarrow \infty} V(t) = 0$

Maks poeng: 10

3 2024 APPC a)

a)

Consider the system

$$y_p = \frac{Z_p(s)}{R_p(s)} u_p$$

where $Z_p(s) = 1$ and $R_p(s) = s - 1$. A pole-placement controller has been designed for this system, placing the closed loop poles at $A^* = (s + 1)^3$, and aiming at tracking the reference y_m asymptotically in time. The result is characterized by the polynomials $L(s) = 1$ and $P(s) = 4s^2 - s + 5$. Which of the following reference signals, if any, will the closed loop system be able to track (ie $y_p \rightarrow y_m$ for $t \rightarrow \infty$)?

Select one or more alternatives (wrong selections give negative score, leaving all boxes unchecked gives zero points).

- | | |
|--|---|
| ☐ $y_m = 0$ | ✓ |
| ☐ $y_m = 3\cos(2t) - \sin(t)$ | |
| ☐ $y_m = 2\sin(t) - 4\cos(t)$ | |
| ☐ $y_m = 2\cos(2t) - 3\sin(2t)$ | ✓ |
| ☐ $y_m = \frac{1}{1+t}$ | ✓ |
| ☐ $y_m = t$ | |
| ☐ $y_m = t^2$ | |
| ☐ $y_m = e^t$ | |
| ☐ $y_m = \sin(2t + 4)$ | ✓ |
| ☐ $y_m = e^{-t}$ | ✓ |

Maks poeng: 15

4 2024 APPC b)

b)

To design an adaptive version of the PPC in a), a parameter estimator is required. Below are four suggestions for obtaining estimates of $\mathbf{a}$ and $\mathbf{b}$ in the system

$$\mathbf{y}_p = \frac{\mathbf{b}}{s+\mathbf{a}} \mathbf{u}_p$$

Which one would you prefer?

Select one (wrong answer gives negative score, while "Don't know" or no answer gives 0 points):

$$\dot{x} = -5x + 5y_p, \quad x(0) = 0$$

$$z = y_p + x$$

$$\dot{\phi}_1 = -5\phi_1 + y_p, \quad \phi_1(0) = 0$$

$$\dot{\phi}_2 = -5\phi_2 - u_p, \quad \phi_2(0) = 0$$

☐
$$\varepsilon = \frac{z - \hat{a}\phi_1 - \hat{b}\phi_2}{1 + \phi_1^2 + \phi_2^2}$$

$$\dot{\hat{a}} = 10\varepsilon\phi_1$$

$$\dot{\hat{b}} = 10\varepsilon\phi_2$$

$$\dot{x} = -5x + 5y_p, \quad x(0) = 0$$

$$z = y_p + x$$

$$\dot{\phi}_1 = -5\phi_1 + y_p, \quad \phi_1(0) = 0$$

$$\dot{\phi}_2 = -5\phi_2 - u_p, \quad \phi_2(0) = 0$$

☐
$$\varepsilon = \frac{z - \hat{a}\phi_1 - \hat{b}\phi_2}{1 + \phi_1^2 + \phi_2^2}$$

$$\dot{\hat{a}} = 10\varepsilon\phi_1$$

$$\dot{\hat{b}} = \begin{cases} 10\varepsilon\phi_2 & \text{for } \hat{b} > 0.1 \text{ or } \varepsilon\phi_2 > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\dot{x} = -5x + 5y_p, \quad x(0) = 0$$

$$z = y_p - x$$

$$\dot{\phi}_1 = -5\phi_1 - y_p, \quad \phi_1(0) = 0$$

$$\dot{\phi}_2 = -5\phi_2 + u_p, \quad \phi_2(0) = 0$$

☐
$$\varepsilon = \frac{z - \hat{a}\phi_1 - \hat{b}\phi_2}{1 + \phi_1^2 + \phi_2^2}$$

$$\dot{\hat{a}} = 10\varepsilon\phi_1$$

$$\dot{\hat{b}} = 10\varepsilon\phi_2$$

☐ Don't know

$$\dot{x} = -5x + 5y_p, \quad x(0) = 0$$

$$z = y_p - x$$

$$\dot{\phi}_1 = -5\phi_1 - y_p, \quad \phi_1(0) = 0$$

$$\dot{\phi}_2 = -5\phi_2 + u_p, \quad \phi_2(0) = 0$$

$$\circ \quad \varepsilon = \frac{z - \hat{a}\phi_1 - \hat{b}\phi_2}{1 + \phi_1^2 + \phi_2^2}$$



$$\dot{\hat{a}} = 10\varepsilon\phi_1$$

$$\dot{\hat{b}} = \begin{cases} 10\varepsilon\phi_2 & \text{for } \hat{b} > 0.1 \text{ or } \varepsilon\phi_2 > 0 \\ 0 & \text{otherwise} \end{cases}$$

Maks poeng: 10

5 2024 APPC c)

c)

Consider the closed loop consisting of the process given in a) in closed loop with the adaptive PPC obtained by using the controller from a) in combination with the parameter estimator from b). For which of the following reference signals, if any, will you expect to achieve both asymptotic tracking (ie $y_p \rightarrow y_m$ for $t \rightarrow \infty$) and parameter convergence (ie $\hat{a} \rightarrow a$ and $\hat{b} \rightarrow b$ for $t \rightarrow \infty$)?

Select one or more alternatives (wrong answers give negative score, while leaving all boxes unchecked gives 0 points).

☐ $y_m = e^{-t}$

☐ $y_m = e^t$

☐ $y_m = 3\cos(2t) - \sin(t)$

☐ $y_m = 0$

☐ $y_m = 2\cos(2t) - 3\sin(2t)$



☐ $y_m = 2\sin(t) - 4\cos(t)$

☐ $y_m = t^2$

☐ $y_m = \frac{1}{1+t}$

☐ $y_m = \sin(2t + 4)$



☐ $y_m = t$

Maks poeng: 10

6 2024 MRAC a)

a) Consider the system

$$y_p = \frac{s+b_0}{(s+a_0)(s+a_1)} u_p \quad (1)$$

where b_0, a_0, a_1 are unknown constants. We would like to design a direct MRAC with un-normalized adaptive laws, for the reference model

$$y_m = \frac{1}{1+s} r \quad (2)$$

where r is a piecewise continuous and bounded reference signal.

What conditions would normally be assumed for system (1)?

Choose one option (wrong answer gives negative score, "Don't know" or no answer gives 0 score).

- ☒ $b_0 > 0$ and $a_0 \neq b_0$ and $a_1 \neq b_0$ ✓
- ☐ $b_0 \neq 0$ and $a_0 \neq b_0$ and $a_1 \neq b_0$
- ☐ $b_0 > 0$ and $a_0 \neq 0$ and $a_1 \neq 0$
- ☐ $b_0 \neq 0$ and $a_0 \neq a_1$
- ☐ Don't know

Maks poeng: 4

7 2024 MRAC b)

IMPORTANT: Below, you will be asked to fill in missing parts of the derivation of the direct MRAC. To make sure your answer is interpreted correctly, DO **NOT** USE SPACES BETWEEN SYMBOLS YOU ENTER.

b) The MRC for system (1) in a) is

$$u_p^* = \theta_1^* \frac{1}{s+f} u_p + \theta_2^* \frac{1}{s+f} y_p + \theta_3^* y_p + r, \quad (3)$$

where $\theta_1^*, \theta_2^*, \theta_3^*$ are constants that depend on the system parameters in (1) and reference model parameters in (2), while f is a constant selected by the designer. The controller can be realized as (fill in the missing parts):

$$\dot{\omega}_1 = \boxed{} (-f) \omega_1 + u_p$$

$$\dot{\omega}_2 = \boxed{} (-f) \omega_2 + y_p$$

$$u_p = \theta_1^* \omega_1 + \theta_2^* \omega_2 + \theta_3^* y_p + r$$

Let (A, B, C) be a realization of system (1), so that

$$\dot{x}_p = Ax_p + Bu_p, \quad y_p = Cx_p.$$

Defining

$$X_p = \begin{bmatrix} x_p \\ \omega_1 \\ \omega_2 \end{bmatrix}$$

the closed-loop dynamics can be expressed as:

$$\dot{X}_c = A_c X_c + B_c r + B_c \tilde{\theta}^T \omega, \quad y_p = C_c X_p$$

where $\tilde{\theta} = [\theta_1 - \theta_1^*, \theta_2 - \theta_2^*, \theta_3 - \theta_3^*]^T$, $\omega = [\omega_1, \omega_2, y_p]^T$ and

$$A_c =$$

$\boxed{} (A) + \theta_3^* \boxed{} (BC)$	$\theta_1^* \boxed{} (B)$	$\theta_2^* \boxed{} (B)$
$\theta_3^* \boxed{} (C)$	$\boxed{} (-f) + \theta_1^*$	θ_2^*
$\boxed{} (C)$	$\boxed{} (0)$	$\boxed{} (-f)$

$$B_c =$$

	(B)
	(1)
	(0)

 $C_c =$

	(C)		(0)		(0)
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The corresponding transfer function from r to y_p is

$$y_p = C_c(sI - A_c)^{-1}B_c r = \left(\frac{\boxed{}(1)}{\boxed{}(s+1, 1+s)} \right) r.$$

It can be proven that the dynamics of the tracking error $e_1 = y_p - y_m$ is governed by

$$\dot{e} = A_c e + B_c \tilde{\theta}^T \omega, \quad e_1 = C_c e,$$

where $e = X_c - X_m$ with X_m the internal states of an appropriate realization of the reference model (2) from a). Consider the Lyapunov function candidate

$$V = \frac{1}{2} e^T P e + \frac{1}{2} \tilde{\theta}^T \Gamma^{-1} \tilde{\theta}.$$

The time derivative of V is (fill in the missing parts, you may omit the subscript c and write A, B, C instead of A_c, B_c, C_c , respectively):

$$\dot{V} = \frac{1}{2} e^T \left(\boxed{} (A_c^T P + P A_c, P A_c + A_c^T P, A^T P + P A, P A + A^T P) \right) \\ e + \tilde{\theta}^T \Gamma^{-1} (\dot{\theta} + \Gamma \omega (P \boxed{} (B_c, B))^T e)$$

Maks poeng: 21

8 2024 MRAC c-d)

(Write on paper, scan and upload).

c) Based on the Lyapunov function and its derivative from b), select an adaptive law and argue that it is implementable.

d) Based on the results from b), and your choice of adaptive law from c), prove that the tracking error $e_1 = y_p - y_m \rightarrow 0$ as $t \rightarrow \infty$ (detailed proof required).

Maks poeng: 10

9 2024 - Adaptive backstepping

Consider the system

$$\dot{x}_1 = x_2 + \theta^* x_1^3$$

$$\dot{x}_2 = x_2 + u$$

where θ^* is an unknown constant parameter and we would like to design u so that the state is regulated to the origin. Applying the standard adaptive backstepping method, using

$q_1 = -cx_1 - \theta x_1^3$ as virtual control, where θ is an estimate of θ^* and $\tilde{\theta} = \theta - \theta^*$, you will arrive at one of the controllers below. Which one (wrong answer gives negative score, while "Don't know or no answer gives 0 points)?

- ☒ $\dot{\theta} = \gamma(x_1^4 + (3\theta x_1^5 + cx_1^3)(x_2 + \theta x_1^3 + cx_1))$
 $u = -(1 + c^2)x_1 - (1 + 2c)x_2 - (2c\theta + \dot{\theta})x_1^3 - 3\theta x_1^2(x_2 + \theta x_1^3)$ ✓
- ☐ $\dot{\theta} = \gamma(x_1^4 - (3\theta x_1^5 + cx_1^3)(x_2 + \theta x_1^3 + cx_1))$
 $u = -(1 + c^2)x_1 - (1 + 2c)x_2 - (2c\theta + \dot{\theta})x_1^3 - 3\theta x_1^2(x_2 + \theta x_1^3)$
- ☐ $\dot{\theta} = \gamma(x_1^4 + (3\theta x_1^5 + cx_1^3)(x_2 + \theta x_1^3 + cx_1))$
 $u = -(1 + c^2)x_1 + (1 + 2c)x_2 - (2c\theta + \dot{\theta})x_1^3 - 3\theta x_1^2(x_2 + \theta x_1^3)$
- ☐ $\dot{\theta} = \gamma(x_1^4 - (3\theta x_1^5 + cx_1^3)(x_2 + \theta x_1^3 + cx_1))$
 $u = -(1 + c^2)x_1 + (1 + 2c)x_2 - (2c\theta + \dot{\theta})x_1^3 - 3\theta x_1^2(x_2 + \theta x_1^3)$
- ☐ Don't know

Maks poeng: 15

2024 Formula Sheet

Useful formulas and results.

Definition 1: $\phi(t)$ is persistently exciting if there exist constants $T > 0, \alpha > 0$ such that $\frac{1}{T} \int_t^{t+T} \phi(\tau) \phi^T(\tau) d\tau \geq \alpha I$ for all $t \geq 0$.

Lemma 1: A function $f : [0, \infty) \rightarrow \mathcal{R}$ which is bounded below and non-increasing has a limit as $t \rightarrow \infty$.

Lemma 2: If $f, \dot{f} \in \mathcal{L}_\infty$ and $f \in \mathcal{L}_2$, then $f(t) \rightarrow 0$ as $t \rightarrow \infty$.

Lemma 3: Given a Hurwitz matrix A , and matrices B and C . If $G(s) = C(sI - A)^{-1}B$ is strictly positive real (SPR), then for any given $Q = Q^T > 0$ there exists a $P = P^T > 0$ such that $A^T P + PA = -Q$ and $PB = C^T$ (recall that all SPR transfer functions are stable and have relative degree 0 or 1. For example, $1/(Ts + 1)$ is SPR for $T > 1$).

Static Parametric Model: $z = \theta^{*T} \phi$

Gradient method with instantaneous cost:

$$\begin{aligned}\dot{\theta} &= \Gamma \varepsilon \phi, \theta(0) = \theta_0 \\ \varepsilon &= \frac{z - \theta^T \phi}{m_s^2}, m_s^2 = 1 + \phi^T \phi\end{aligned}$$

Gradient method with integral cost:

$$\begin{aligned}\dot{\theta} &= -\Gamma(R(t)\theta + Q(t)), \theta(0) = \theta_0 \\ \dot{R} &= -\beta R + \frac{\phi \phi^T}{m_s^2}, R(0) = 0 \\ \dot{Q} &= -\beta Q - \frac{z \phi}{m_s^2}, Q(0) = 0 \\ m_s^2 &= 1 + \phi^T \phi\end{aligned}$$

Least-squares method with forgetting:

$$\begin{aligned}\dot{\theta} &= P\varepsilon\phi, \theta(0) = \theta_0 \\ \dot{P} &= \beta P - P \frac{\phi\phi^T}{m_s^2} P, P(0) = P_0 \\ m_s^2 &= 1 + \phi^T \phi\end{aligned}$$

Model reference adaptive control:

$$\text{Plant: } y_p = k_p \frac{Z_p(s)}{R_p(s)} u_p, (\deg(R_p) = n_p \leq n, \deg(Z_p) = m_p < n_p)$$

$$\text{Reference model: } y_m = k_m \frac{Z_m(s)}{R_m(s)} r, (\deg(R_m) = p_m, \deg(Z_m) = q_m < p_m)$$

$$\text{Control law: } u_p = \theta_1^{*T} \frac{\alpha(s)}{\Lambda(s)} u_p + \theta^{*T} \frac{\alpha(s)}{\Lambda(s)} y_p + \theta_3^* y_p + c_0^* r,$$

where $\alpha(s) = [s^{n-2}, s^{n-1}, \dots, s, 1]^T$ for $n \geq 2$, $\alpha(s) = 0$ for $n = 1$. $\Lambda(s)$ is monic Hurwitz of degree $n - 1$. The controller parameters satisfy, for all s ,

$$c_0^* = k_m/k_p \text{ and } (\Lambda - \theta_1^{*T} \alpha) R_p - k_p Z_p (\theta_2^{*T} \alpha + \theta_3^* \Lambda) = Z_p \Lambda_0 R_m \text{ where } \Lambda = \Lambda_0 Z_m.$$

$Z_p, R_p, Z_m, R_m, \Lambda, \Lambda_0$ are monic.

Pole placement control:

$$\text{Plant: } y_p = \frac{Z_p(s)}{R_p(s)} u_p, (\deg(R_p) = n, \deg(Z_p) < n)$$

$$\text{Controller: } u_p = \frac{\Lambda - LQ_m}{\Lambda} u_p - \frac{P}{\Lambda} (y_p - y_m)$$

$\deg(Q_m) = q, \deg(P) = q+n-1, \deg(L) = n-1, \deg(\Lambda) = n+q-1$. R_p, Q_m, L, Λ are monic. The controller parameters satisfy, for all s , $Q_m y_m = 0$, $LQ_m R_p + PZ_p = A^*$, where A^* is monic Hurwitz.

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Normal

-Kandidaten må bruke og begrunne bruken av MKY Lemma (står i formelark).

-Kandidaten må skrive opp adaptiv lov på form som er implementerbar (e_1 må brukes og ikke PBe).

-Kandidaten må fastslå at Lyapunov-analysen gir $e, \theta \in \mathcal{L}_\infty$, og $e \in \mathcal{L}_2$.

-Kandidaten må begrunne hvorfor $e \in \mathcal{L}_\infty$ impliserer at $X_c \in \mathcal{L}_\infty$ (nevne at $X_m \in \mathcal{L}_\infty$).

-Kandidaten må argumentere for at de/dt er begrenset.

-Kandidaten må bruke Barbalat's Lemma for å fastslå konvergens av e (står i formelark).

-Kandidaten må si at konvergens av e medfører konvergens av e_1 .

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Oppdater

-Kandidaten må bruke og begrunne bruken av MKY Lemma (står i formelark).

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